

Problem 6.18

Complete the following table.

	Simple interest	Compound interest
$a(t) =$	$1 + it$	$(1 + i)^t$
$a(0) =$	1	1
$a(1) =$	$1 + i$	$1 + i$
Period during which $a(t)$ is greater	$0 < t < 1$	$t > 1$
$i_n =$	$\frac{i}{1 + i(n - 1)}$	i

For simple interest, the accumulation function is $a(t) = 1 + it$, while for compound interest it is $a(t) = (1 + i)^t$.

Also, since both accumulation functions agree at $t = 0$ and $t = 1$, simple interest is greater for $0 < t < 1$, and compound interest is greater for $t > 1$.

Finally, under simple interest the effective rate in the n th period is

$$i_n = \frac{a(n) - a(n - 1)}{a(n - 1)} = \frac{i}{1 + i(n - 1)},$$

whereas under compound interest it is constant and equal to i .